

CNG Cylinder Process Tracking

Workflow Implementation Report

How every cylinder's 21-stage journey is captured, evaluated, stored and delivered — machine to dashboard.



CAPTURE

SAP · machines · operators push every stage event



EVALUATE

each reading checked against stage limits



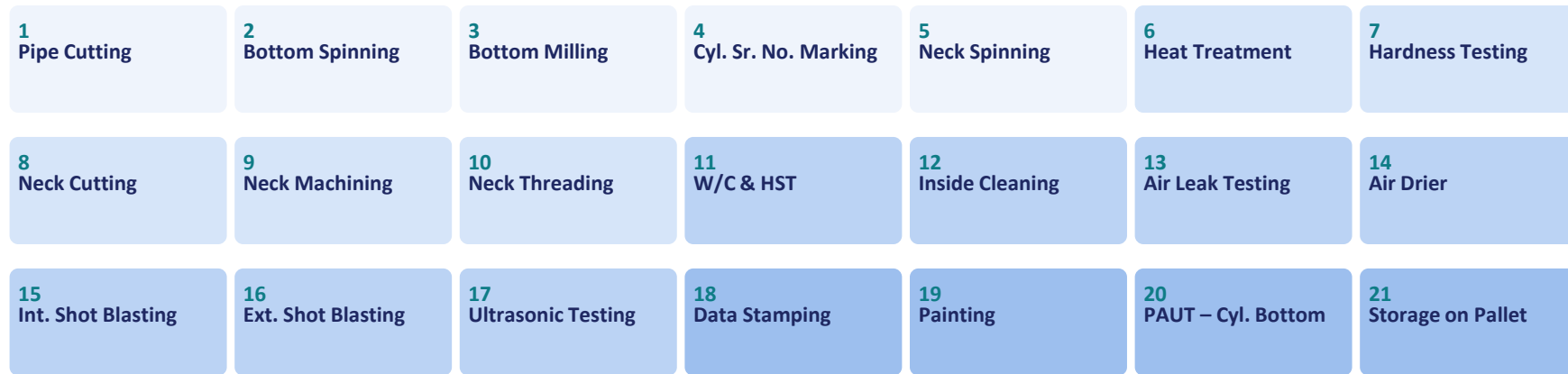
DELIVER

live dashboards, trace & audit for every Pipe ID

THE WORKFLOW

One cylinder, 21 tracked stages

The exact manufacturing flow from your Process Flow document — every stage becomes a digital checkpoint.

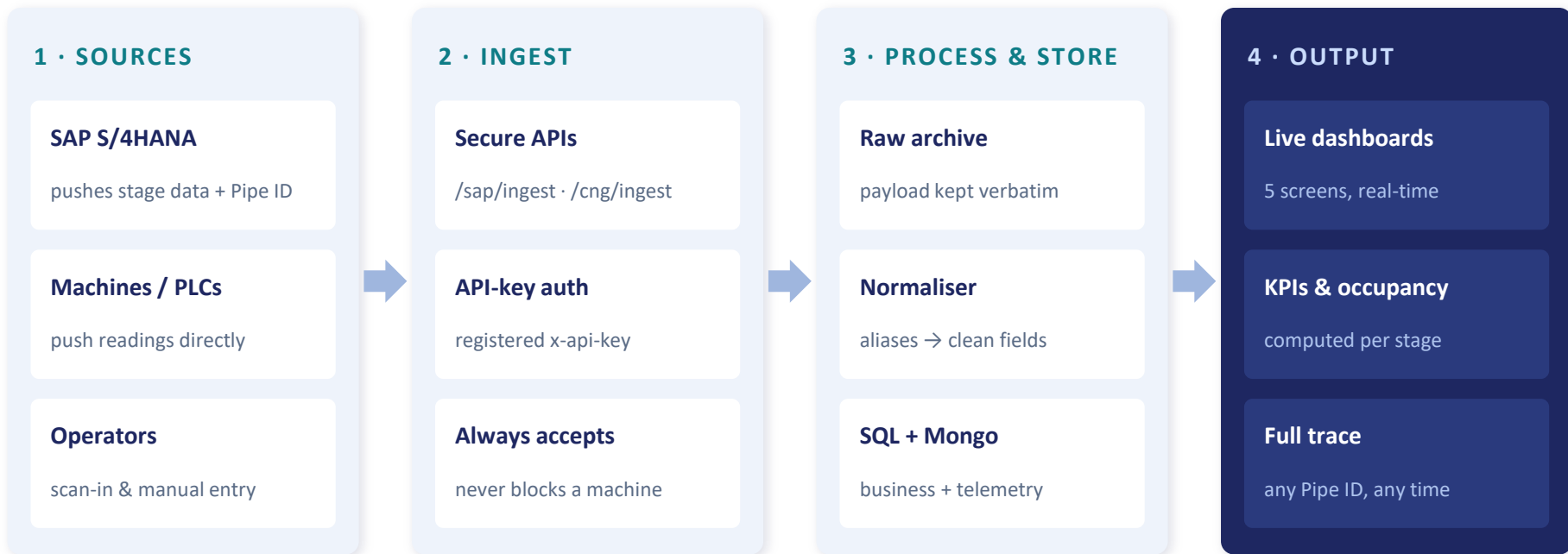


☐ Forming ☐ Heat & Machining ☐ Clean & Test ☒ Finish & Store



Stage 1 gives every cylinder its identity — the Pipe ID. Entered / scanned at Pipe Cutting, it travels through all 21 stages: every machine reading, result and timestamp is written against it. Stages 2–21 are machine-fed automatically.

End-to-end: machine signal to live dashboard



Design rule: capture first, interpret second. The line never waits on software — data is accepted in any shape, archived raw, then normalised and evaluated.

Three ways stage data reaches the system



SAP push (primary)

Minda's SAP maps its data to our published format and POSTs one message per machine event — always carrying the Pipe ID, stage and readings.

`POST /api/v1/sap/ingest`



Machine / PLC push

Machines or edge gateways post JSON directly, exactly like the proven EKC & JCI telemetry lines already running on this platform.

`POST /api/v1/cng/ingest`



Operator entry

Stage 1 (Pipe Cutting) births the Pipe ID via manual entry / barcode scan; scan-in links a cylinder to a machine wherever needed.

`Operator screen / scanner`

ONE STAGE EVENT — THE MESSAGE

```
{
  "machineId": "HT-401",
  "stageNo": 6,
  "pipeId": "JPM24C08921",
  "ts": "2026-07-04T09:41:22Z",
  "data": {
    "furnace_temp": 892,
    "soak_min": 42,
    "result": "OK",
    "status": "run"
  }
}
```

pipeId on every message = traceability with no extra hardware.

Accept everything. Archive raw. Then make it clean.



1 • ACCEPT

Any JSON shape is taken with a 2xx — single object, array batch, nested or flat. A schema change upstream never stops the line.



2 • ARCHIVE RAW

The exact payload is stored untouched (SAP inbox / telemetry store) — a permanent audit copy that can be re-processed anytime.



3 • NORMALISE

A tolerant mapper resolves field aliases (machineId / machine_id / device...), units and timestamps into one clean stage-event record.



4 • VALIDATE

PLC fault sentinels (-32768, 32767, 65535) are quarantined, duplicates dropped, and only clean values reach evaluation.

Why raw-first? Machine formats drift, SAP mappings evolve, vendors change firmware. Because the original payload is always kept, nothing is ever lost — history can be re-normalised under new rules without asking anyone to resend data.

Every reading judged against the stage's limits

1 Range check

Each parameter is compared to the stage's engineering window (min–max per metric — temperature, pressure, hardness, thickness...).

2 Stage result

The stage event is stamped OK or NG and written to the cylinder's permanent stage record with machine, time and values.

3 Defect capture

An NG result opens a defect entry — stage, machine, parameter and value — feeding the Defects board until closure.

4 Machine status

Every message also updates the machine's live state: run / idle / stopped / maintenance / fault, shown on the Machines wall.

AS SEEN ON THE DASHBOARD



Heat Treatment · HT-401

892 °C

window 850 – 950 °C → OK



Air Leak Test · ALT-1301

0.34 bar drop

limit ≤ 0.10 bar → NG · defect raised

Two-tier storage: business truth + raw telemetry



SQL SERVER (JPMDO) — the business record

`cng_cylinder` — one row per Pipe ID — the cylinder master

`cng_stage_record` — one row per stage event: values, result, machine, time

`cng_machine` — live state per machine: status, latest gauges, ranges

`cng_defect` — every NG with stage, parameter and closure state

`sap_inbox` — raw SAP payload archive — full audit trail

`cng_ingest_key` — registered, hashed API keys per data source

Additive schema — new `cng_*` tables only. Existing Minda tables untouched, zero regression risk.



MONGODB — raw telemetry

Every machine packet, verbatim, as a time-series.

Loose schema by design — new sensors and fields appear without any migration.

Feeds re-processing, deep dives and future analytics (OEE, SPC, ML).

Why two tiers? Clean relational data gives fast, reliable dashboards; the raw archive guarantees nothing is ever lost. Each does what it's best at.

From stage records to live answers

Stage occupancy

cylinders currently sitting at each of the 21 stages — the live line-balance picture

WIP / completed

counts derived from each cylinder's furthest stage record — no manual tallies

Quality & defects

OK/NG rates per stage & machine; open defect ageing until closure

Machine liveness

latest reading vs range per gauge; stale feed detection via last-seen time

FIVE LIVE SCREENS



Live Ops

KPIs, occupancy, live event feed



Cylinder Trace

search a Pipe ID → its full 21-stage journey



Machines

machine wall with live gauges & status



History

filter any period, machine or stage

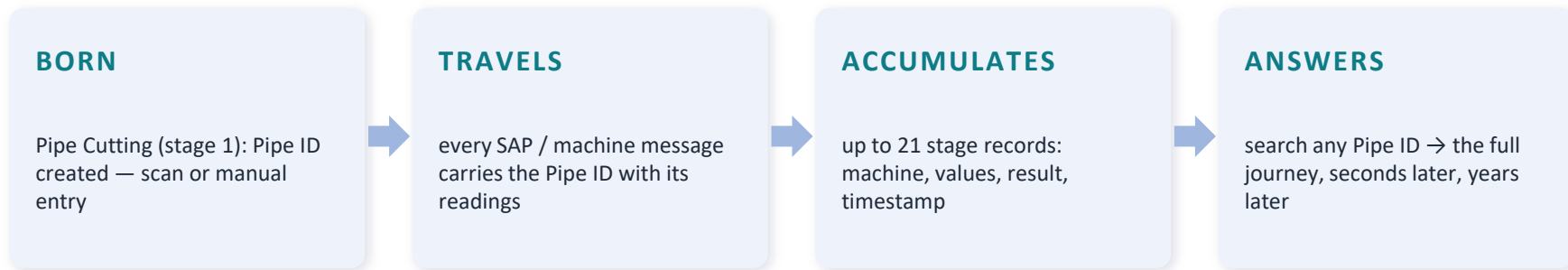


Defects

open NGs, ageing and closure

Real-time by design: Socket.io pushes an update the moment data lands; polling guarantees freshness even on flaky networks.

One Pipe ID — the complete life of a cylinder



1 Pipe ID
= 21 stage records, one audit trail

WHAT THIS UNLOCKS

- Customer & IATF audits answered from a screen, not a paper hunt
- Recall scope narrowed to exact cylinders, batches and machines
- NG root-cause: see the same stage across every affected cylinder

The stack behind it — and why each piece



Frontend **React 19 + Vite + Tailwind**

fast plant-floor dashboards; TanStack Query keeps data fresh



Backend **Node.js · Express 5**

one API layer for ingest, evaluation and dashboards



Business data **SQL Server (Sequelize)**

lives inside Minda's existing JPMDO — one platform, one login



Raw telemetry **MongoDB time-series**

schema-loose archive of every packet, ready to re-process



Real-time **Socket.io + Redis**

instant dashboard updates; cache & session backbone



Security **JWT + x-api-key**

role-based viewing; registered keys for every data source



SAP integration **Push to our spec + OData**

SAP maps to our format; material master synced from S/4HANA



Deployment **Existing Minda infra**

additive module on digitisationapi.jpmgroupp.co.in · PM2

Locked down, auditable, zero-regression



Registered API keys

every pushing system gets its own key — stored hashed, revocable per source, no shared secrets



Role-based dashboards

viewing sits behind the existing Minda login (JWT) and role permissions — same accounts, same RBAC



Immutable raw audit

every inbound payload is archived verbatim with source & timestamp — a tamper-evident record of what was sent



Additive by design

only new `cng_*` tables; no foreign keys into and no changes to existing Minda tables — the platform you run today is untouched



HTTPS end-to-end

all traffic rides the hosted TLS domain; dev, test and production environments fully isolated

WHERE WE ARE

Built, proven on live pushes — ready to scale to the line

DELIVERED & VERIFIED

- Ingest APIs live — machine & SAP pushes verified end-to-end
- Batch + single message formats, dividing into machine / cylinder / defect data
- All 5 dashboard screens on real data, with live gauges & filters
- 16-machine demo line streaming (12 via SAP format + EKC/JCI patterns)
- Published integration spec shared for SAP field mapping

NEXT — TO GO LIVE

- SAP team maps live fields to the published format (sample + key already shared)
- Production API keys issued per source system
- Operator scan-in station at Pipe Cutting (stage 1)
- Stage limit sign-off with quality team (min–max per parameter)
- Go-live line by line, monitored against the raw archive